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Early but discontinued exposure to a language exerts lasting effects on white matter architecture in the brain

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Language experiences influence later cognitive development, but how they shape brain organization remains unclear. We investigated four groups of participants who differed in their early linguistic environments: International Adoptees (IA) from China, who discontinued their first language (L1) Mandarin in the first two years of life and only speak French since adoption; L1-Mandarin speakers with early acquisition and continued use of French; L1-French speakers; and English-French bilinguals. We used diffusion-weighted imaging to explore group differences in two white-matter tracts in the brain that play a role in language processing, the arcuate fasciculus and superior longitudinal fasciculus. Our findings revealed that despite discontinuing Mandarin early in life, IA participants exhibited structural features resembling those of native Mandarin speakers, including smaller tract volumes in the left hemisphere and more balanced tract volumes across hemispheres. In contrast, English-French bilingual and L1-French groups exhibited similar white-matter characteristics, which were distinct from the two tone-language exposed groups, suggesting a structural organization specific to tonal language experience, and not bilingualism in general.

A fundamental aspect of unravelling the puzzle of language acquisition is the question of what is universal under all circumstances of language learning, and how the brain is impacted by the specific characteristics of the languages being learned. The human brain undergoes significant changes during development, with white-matter tracts playing a critical role in connecting different cortical regions and supporting cognitive functions. Among the most important of these tracts is the arcuate fasciculus (AF)[1,2]. The AF connects posterior temporal areas (i.e., Wernicke's area) with the inferior frontal areas (often referred to as Broca's area) along the dorsal language stream[2]. Especially in the language-dominant left hemisphere, the role of the AF in language has been described as largely being important for mapping sound to articulation but also, because of its bidirectionality, as being involved in top-down attentional aspects of auditory processing[1].

Although left lateralization for language in the brain is already present in early childhood[3–6], language lateralization is thought to develop from an almost equipotential state that exists in the first two years of life, with a progressive loss of this equipotentiality occurring as linguistic skills come under the predominant control of one side of the brain[7]. Developmental changes in lateralization of language networks may occur partly because of brain maturation, but some changes in lateralization are driven by specific language experiences. For example, pitch processing is typically right-lateralized, but becomes left-lateralized in native tonal language speakers[8,9]. In tonal languages, pitch contours are integral to distinguishing word meanings[10]. Thus, for native-tonal language speakers when pitch carries necessary linguistic information, it is processed in the left hemisphere. However, when pitch carries primarily paralinguistic contextual information (without any meaning/semantic information), such as speaker gender or emotion, it is processed in the right hemisphere. Research using functional neuroimaging has shown that due to the importance of processing and integrating pitch with linguistic information, speakers of tonal languages exhibit increased activation in the right frontal and temporal auditory cortex compared to speakers of non-tonal languages[11–15], highlighting the bilateral engagement of brain regions during speech perception of tonal languages[16]. The AF's involvement in phonological processing[2,17–19] makes it a key player in tone processing, particularly in the early phases of the acquisition of

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tonal languages, such as Mandarin. In early childhood, the AF is involved in mapping auditory input to motor output, facilitating the child's ability to reproduce the phonetic variations they hear. This process is crucial for language development, as children need to both perceive and produce the tonal differences that convey meaning in their native language. Moreover, the first year of life is a crucial period (closing around 10–12 months of life) within which language-specific perceptual development occurs, leading to the establishment of essential language-specific perceptual units[20].

In addition to the AF, the third branch of the superior longitudinal fasciculus (SLF III) is also important for the processing of speech sounds, especially in the context of speech production and articulation[21–24]. The left SLF III connects the inferior frontal areas with inferior parietal areas, more specifically the supramarginal gyrus, which is known to play a role in phonological processing, and by connecting with frontal areas, creates a phonological or articulatory loop for speech production[1]. The SLF III also has connections with areas of the middle frontal gyrus (area 9/46 v) involved in the monitoring of information in working memory[25] and is thus suggested to play a role in the monitoring of phonological information and speech[23,24]. Because of its role in phonological processing, the SLF is likely also affected by variations in early exposure to the sound systems of different languages.

The AF and SLF III work together to support language processing, but they each play specific roles given that the cortical areas they connect differ. These tracts undergo significant development in early childhood, a period marked by rapid language acquisition. Diffusion imaging studies, for example, have shown that the AF and SLF become increasingly myelinated during the first few years of life[26–29], reflecting the child's growing capacity for language processing and phonological discrimination. Especially at a young age, language experience has been shown to have a strong effect on how brain networks develop[30]. What is not known is precisely how brain development may be sensitive to the specific properties of the language(s) to which the developing child is exposed.

Early developmental experiences play a crucial role in shaping later neural outcomes, especially as children are exposed to an increasingly diverse range of linguistic environments. It has long been suggested that the most rapid pace of learning takes place during the first years of life and that during this time of heightened neuroplasticity, the brain is optimally predisposed to collect and store basic information about the world[31]. Hearing a language during early development helps infants' brains become attuned to the specific sounds of that language, forming neural representations of these sounds[32,33]. These neural representations serve as a foundational base for learning increasingly complex and structured aspects of the language[34]. This suggests that there is a continuous connection between the neural representations established early on and the higher-level language abilities that develop over the following years. The adaptive importance of this relationship is that it enables the developing individual to build more advanced language skills upon this early neural foundation. However, there is a lack of substantial empirical evidence regarding how these early experiences influence neural processing in later stages of development. In a recent publication, Pierce et al.[35] showed evidence for the maintenance of neural templates for the processing of Chinese tones in a group of adolescent international adoptees (IA) whose exposure to their birth language (Chinese) was totally discontinued when they were 12.8 months of age, on average. From that point on, the IAs were exposed to and spoke only French and had no conscious recollection of their birth language when tested more than a decade later. However, when these participants listened to a tone that carried linguistic information present only in Chinese, and not in their current language (French), their pattern of brain activation matched that observed in native Chinese speakers who had spoken Chinese continuously since birth and had acquired French as a second language at approximately the same age as the IA. Importantly, the pattern of activation demonstrated by the IA differed significantly from that of French L1 speakers who had never been exposed to Chinese, even though all participants heard identical acoustic stimuli. The fact that the neurocognitive responses of the adopted participants closely matched those of the native Chinese speakers provided evidence that the neural representations supporting the processing of Chinese had been acquired during the first months of life and were not overwritten or lost over time, but were maintained in the brain. Pierce et al.[36] also showed, in a subsequent paper, that these traces in the adoptees affected later language processing of their adoptive language, French.

The present paper builds on and extends this prior research that examines the functional outcomes of early language exposure on brain function by examining the impact of language exposure on brain anatomy, specifically, structural differences in the white matter tracts involved in processing language sounds. We examined in what way the functional changes observed in temporal auditory and language processing areas are underpinned by the white matter tracts connecting these regions to frontal language areas—specifically the AF and the SLF III. We thus used anatomical diffusion imaging tractography to shed light on how linguistic experience shapes the brain structurally. We compared four participant groups that were exposed to tonal and non-tonal languages to shed light on how variation in first-language exposure shapes the brain: Chinese International Adoptees (IA) who stopped speaking Mandarin in the first two years of life and now speak only French; Mandarin L1 speakers with early, ongoing French exposure (i.e., Mandarin-French bilinguals; Mandarin group); French monolinguals; and English-French bilinguals. We asked whether brain structure is language-independent, in which case IA participants—who ceased using Mandarin shortly after adoption- should converge structurally with monolingual French speakers. Alternatively, if early sensory and linguistic input largely shape neural architecture during a sensitive developmental window[37,38], then the white-matter profiles of the IA should be more aligned with those of tone-language speakers, i.e., the Mandarin group.

Results

The left and right AF and SLF III were successfully dissected in all participants. Table 1 displays the group mean values of volume and fractional anisotropy (FA) for each tract and Fig. 1 shows an example of the dissected tracts in one participant.

Group differences

The four groups did not differ in terms of age ($F = 1.71$, $p = 0.169$), although the E/F bilingual group was older overall (Range: 18–25, $M = 21.6$). The primary purpose of including the English/French (E/F) bilingual group was to determine whether the pattern observed in the Mandarin bilingual group was specifically related to tone language exposure, rather than bilingualism in general. The two bilingual groups were matched in terms of age of acquisition of their second language or the number of years of experience with the second language (see methods section for statistics).

To investigate group differences in terms of white matter tract volumes, an initial linear mixed effects model testing the impact of Hemisphere (left, right), Tract (AF, SLF), Group, and Age as interaction factors (with Study as a control factor) revealed a significant Tract by Age interaction ($t = -3.435$, $p = 0.0007$) with the AF showing greater volume increases with Age compared to the SLF III (See supplementary table 1 for detailed statistical results). Since no higher order interactions, including the Age factor reached significance ($|t| < 1.105$, $|p| > 0.269$), a follow-up model was fitted including Age and Study as control factors. This model revealed significant Hemisphere by Group interactions, where the French group ($b = -3.058$, $SE = 1.918$, $t = -2.205$, $p = 0.0281$) and the E/F bilingual group ($b = -5.361$, $SE = 1.498$, $t = -3.578$, $p = 0.0004$) showed greater hemispheric asymmetry for both tracts compared to the IA group used as reference. The same interaction failed to reach significance for the Mandarin group compared to the IA group ($b = -0.966$, $SE = 1.481$, $t = -0.652$, $p = 0.515$). Visual inspection of the data (Fig. 1) suggests that, for the AF and SLF, the French and E/F bilingual groups had larger tract volumes than the IA and Mandarin group on the left but not the right hemisphere. The model yielded a significant interaction between Tract and Group only for the Mandarin group compared to the IA group ($b = -3.065$, $SE = 1.481$, $t = -2.070$, $p = 0.039$).

Table 1 | Participants' characteristics, mean (*SD* in parentheses) fractional anisotropy (FA) and volumes of the arcuate fasciculus (AF) and superior longitudinal fasciculus (SLF III) for each group

			French	IA	Mandarin	E/F bilinguals
	n		33	36	26	25
	Age (years)		19.8 (6.0)	18.5 (6.7)	18.7 (5.7)	21.5 (1.9)
	French Duration (years)		19.8 (6.0)	17.3 (6.4)	15.0 (4.5)	21.5 (1.9)
AF	Left	Volume (mL)	51.20 (5.7)	46.10 (7.1)	48.33 (5.9)	52.58 (5.8)
		FA	0.459 (0.02)	0.460 (0.02)	0.457 (0.02)	0.484 (0.02)
	Right	Volume (mL)	45.81 (9.0)	43.47 (6.7)	45.96 (7.0)	46.29 (8.2)
		FA	0.439 (0.02)	0.441 (0.02)	0.440 (0.02)	0.450 (0.02)
SLF III	Left	Volume (mL)	24.05 (5.4)	19.76 (5.6)	20.15 (4.8)	26.66 (5.0)
		FA	0.442 (0.02)	0.443 (0.02)	0.440 (0.02)	0.458 (0.02)
	Right	Volume (mL)	26.90 (5.8)	25.96 (4.6)	24.16 (6.1)	25.80 (7.7)
		FA	0.449 (0.02)	0.447 (0.02)	0.444 (0.02)	0.453 (0.20)

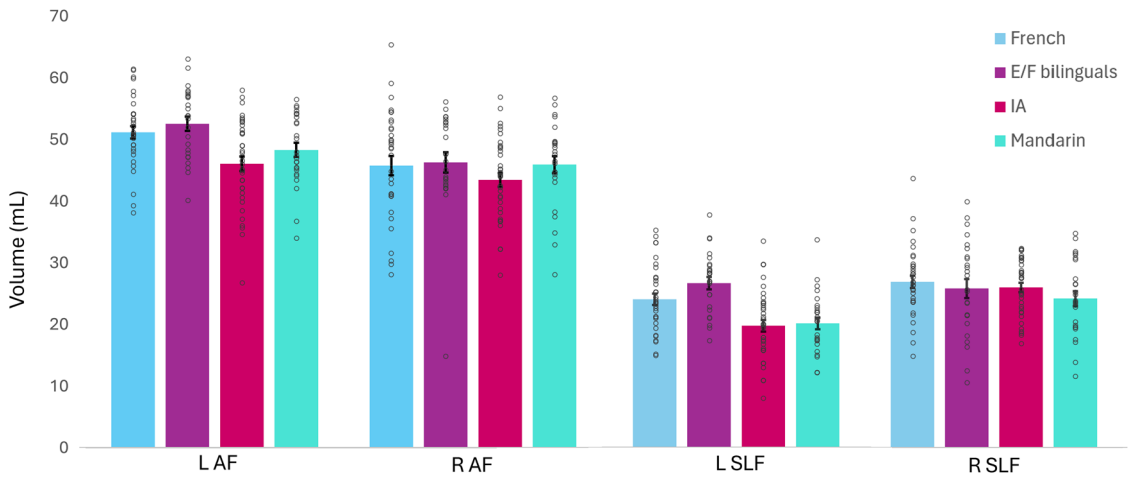

Fig. 1 | Volumes of the left and right arcuate fasciculus (L AF, R AF) and superior longitudinal fasciculus (L SLF, R SLF) for each group. Black circles represent individual datapoints, and error bars represent the standard error of the mean.

The same interaction failed to reach significance for the French group and the E/F bilingual group ($|b| < 1.276$, $|SE| > 1.387$, $|t| < 0.852$, $|p| > 0.385$) compared to the IA group. The LME model also yielded a significant Hemisphere by Tract interaction ($b = 8.837$, $SE = 1.918$, $t = 4.607$, $p < 0.0001$), with larger white matter volumes in the left hemisphere than in the right hemisphere for the AF but not for the SLF III. No higher order interactions including Hemisphere, Tract and Group reached significance ($|b| < 3.419$, $|SE| > 2.773$, $|t| < 1.141$, $|p| > 0.255$), see supplementary table 2 for complete model output.

The same analysis was conducted with tract fractional anisotropy (FA) measures instead of volume. The first model with age as an interaction factor revealed a four-way interaction (Group by Tract by Hemisphere by Age) which was only significant for the E/F bilingual group compared to the IA group ($b = -5.328e{-}03$, $SE = 2.686e{-}03$, $t = -1.984$, $p = 0.0481$). There was no other interaction with Age so like for volume, the model was fitted with Age and Study as control factors. There were significant Hemisphere by Group interactions for both tracts, where only the E/F bilingual group showed greater hemispheric asymmetry compared to the IA group used as reference ($b = -1.14e{-}02$, $SE = 3.15e{-}03$, $t = -3.631$, $p = 0.0003$). The same interaction failed to reach significance for the Mandarin ($b = 6.62e{-}{-}04$, $SE = 3.11e{-}03$, $t = 0.213$, $p = 0.8214$) or French group ($b = 1.87e{-}03$, $SE = 2.91e{-}03$, $t = 0.642$, $p = 0.5215$) compared to the IA group. The model yielded a significant interaction between Tract and Group only for the E/F bilingual group compared to the IA group ($b = -6.84e{-}03$, $SE = 3.15e{-}03$, $t = -1.176$, $p = 0.0302$). The same interaction failed to reach significance for the French group and the E/F bilingual group ($|b| < 2.07e{-}03$, $|SE| > 2.91e{-}03$, $|t| < 0.711$, $|p| > 0.4775$) compared to the IA group. The LME model also yielded a significant Hemisphere by Tract interaction ($b = 2.17e{-}02$, $SE = 4.03e{-}03$, $t = 5.380$, $p < 0.0001$), with larger FA values in the left hemisphere than in the right hemisphere for the AF but not for the SLF III. No higher order interactions, including Hemisphere, Tract and Group reached significance ($|b| < 7.13e{-}03$, $|SE| > 5.82e{-}03$, $|t| < 1.134$, $|p| > 0.2576$), see supplementary tables 3 and 4 for complete model outputs.

To ensure that the observed group differences were not driven by the E/F bilingual group, which included older (i.e., only adult) participants from a separate study, and because age was a significant factor in some analyses, we ran the analysis again with only the IA, Mandarin, and French groups.

For tract volume, there were significant Hemisphere by Group interactions for both tracts, where only the French group showed greater hemispheric asymmetry compared to the IA group used as reference ($b = -3.058$, $SE = 1.3086$, $t = -2.337$, $p = .0202$). The same interaction failed to reach significance for the Mandarin group ($b = -0.9656$, $SE = 1.3975$, $t = -0.691$, $p = 0.4902$) compared to the IA group. The model yielded a significant interaction between Tract and Group only for the Mandarin group compared to the IA group ($b = -3.0649$, $SE = 1.3975$, $t = -2.193$, $p = 0.0291$). The same interaction failed to reach significance for the French

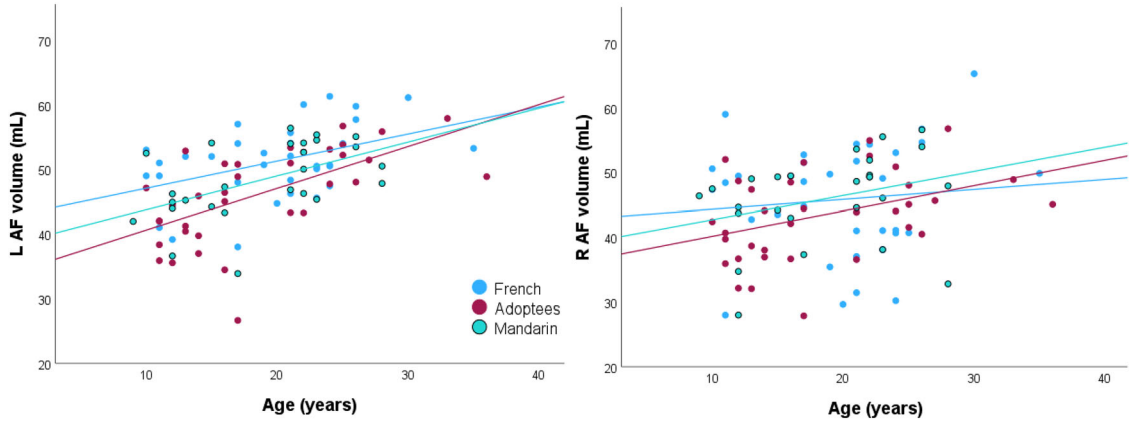

Fig. 2 | Relationship between age and left (L) and right (R) arcuate fasciculus (AF) volumes in each group.

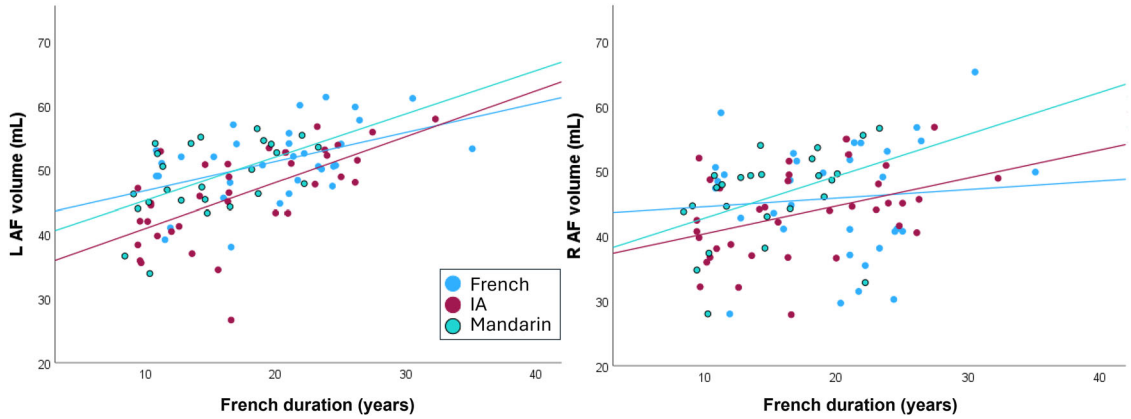

Fig. 3 | Relationship between duration of French exposure (number of years since they acquired French) and left (L) and right (R) arcuate fasciculus (AF) volumes in each group.

group ($b = -1.1061$, $SE = 1.3086$, $t = -0.845$, $p = 0.3987$) compared to the IA group. The rest of the results were similar to those with four groups, see Supplementary table 5.

For FA, there was no significant interaction with Group, only a significant Hemisphere by Tract interaction ($b = 2.17e{-}02$, $SE = 3.79e{-}03$, $t = 5.724$, $p < .0001$). See Supplementary Table 6 for detailed results.

Relationship with chronological age

Because white matter structure in the brain changes throughout development—and given the wide age range among participants in three of the four groups, as well as the fact that some individuals were tested many years after adoption—we examined how chronological age affected tract properties. The E/F bilingual group was excluded from this analysis, as it consisted entirely of adult participants with little variability in terms of chronological age compared to the other groups.

The initial LME model, including age as an interaction term, yielded a significant interaction between Tract and Age; we thus explored the effect of age in each tract separately. For the volume of the left AF, there was a Group by Age Interaction ($F = 12.37$, $p < 0.001$) where the volume of the tract increases less with age in the French group (it starts off larger in the youngest participants compared to the other groups). This also means that the group difference observed for the left AF was mostly driven by the younger individuals. There was no Group by Age interaction for the right AF: $F = 2.262$, $p = 0.087$, the left SLF III: $F = 0.396$, $p = 0.756$, or the right SLF III: $F = 1.611$, $p = 0.192$. Figure 2 displays the relationship between age and AF tract volumes in each group.

Duration of French exposure

This relationship between age and tract volume was further investigated by examining the effects of duration of exposure to French (how many years since they acquired French, which is the same as chronological age for the French group ($M = 19.5$, $SD = 6.5$) but not for the Mandarin ($M = 15.0$, $SD = 4.5$) and IA ($M = 16.7$, $SD = 6.4$) groups) on AF volumes. There was a Group by French Exposure Duration interaction for the L AF ($F = 14.084$, $p < 0.001$) and the R AF ($F = 2.943$, $p = 0.037$) but not for the left ($F = 0.650$, $p = 0.585$) or right ($F = 2.295$, $p = 0.083$) SLF III.

The L AF volume correlates with Age (Fig. 2) and duration of French exposure (Fig. 3) in all groups, but the R AF correlates with Age ($r = 0.390$, $p = 0.019$) and duration of French exposure only in the IA group ($r = 0.408$, $p = 0.013$). In the Mandarin group, duration of French exposure (but not age) correlates with the R AF volume ($r = 0.411$, $p = 0.041$). This means that in the IA and Mandarin groups only, more years of experience with French is linked with larger left and right AF, while in the French group, only the left AF volume increases with years of experience.

In the IA group, age of adoption was not related to tract volumes (L AF: $r = -0.152$, $p = 0.377$; R AF: $r = -0.164$, $p = 0.339$; L SLF III: $r = 0.06$, $p = 0.723$; R SLF III: $r = -0.053$, $p = .759$) but "duration of French exposure", corresponding to how long ago they were adopted (and also related with chronological age), was an important predictor for both left and right AF development.

It appears that it is not the age of adoption that has an effect on tract volume, but longer experience with the new language, French. Over time, IA seems to keep following a similar developmental trajectory as the Mandarin group, with both hemispheres developing in parallel, while the "French

brain" remains left-lateralized. With age, the group difference attenuates. This suggests that due to early tonal language experience, the white matter tracts (AF) in the left and right hemisphere develop more similarly, but if instead a child is only exposed to French (or a similar non-tonal language), the brain develops to rely more on the left AF than the right AF.

Discussion

Findings from animal work suggest that neural maps that are established early have an 'enduring trace' (e.g., auditory system in barn owls[39], visual system in mice[40]). We know that input from the outside world can shape and influence the human brain; a more contentious issue is whether such changes endure without continued input[41–43]. Moreover, the specific properties of languages to which individuals are exposed during development may affect brain development differently. The present study addressed these questions in a unique population of language learners--international adoptees (IA)--and compared them to other language learners. Using diffusion imaging tractography, we show that IA from China, exposed mainly to French since adoption in the first two years of life, maintained the neural architecture of their birth language despite no ability to use that language and no conscious recollection of it. Thus, the IA, who are functionally monolingual French speakers, demonstrate white matter tracts connecting the language areas of the brain that are more similar to a group of Mandarin-L1 speakers than to groups of French-L1 speakers or English/French (E/F) bilinguals. Especially in the language-dominant left hemisphere, the IA have smaller arcuate fasciculus (AF) volumes than the French and E/F bilingual groups but do not differ from the Mandarin group. The French and E/F bilingual groups (i.e., participants who were never exposed to tonal languages) show the classic strong left-lateralization for this tract[1,44–46], while the other two groups display a more bilateral distribution with a shift away from the left. Similar results are observed for the other language-related tract, the superior longitudinal fasciculus (SLF III), where the French group has larger volumes than the two groups exposed to tonal languages in the left but not the right hemisphere. Looking at relationships with age revealed patterns specific to the development of the AF, which were not observed for the SLF III. In the French group, only the left AF showed a significant increase in volume with age (in the age range 9–35 years), while both the left and right AF volume increased with age in the IA group. Interestingly, the properties of the AF in the IA were not affected by the age of adoption (or how long they were exposed to the tonal language), but rather by how long they were exposed to their new primary language, French.

The purpose of including an E/F bilingual group was to make sure that the structural differences observed were indeed due to tonal language exposure and not to bilingualism in general. This group showed a pattern of brain anatomy that was very similar to the French-speaking group and differed from the Mandarin and IA groups. Consistent with our results, Vaquero et al.[47] observed that sequential English-French bilinguals showed the same classic left-lateralization as in our E/F bilingual group and also observed that enhanced left-ward lateralization of the AF volume was linked with bilingualism (of two non-tonal languages) specifically in individuals who learned their second language together with the first, from birth. Moreover, Hämäläinen et al.[48] observed, in bilinguals who spoke two non-tonal languages, that the properties of the left AF were modulated by higher phonological processing load associated with language learning early in life. However, Cui et al.[49] did not find any link between multilingual experience specifically and the properties of the AF. On the other hand, early musical experience and musical training, which can be related to tone perception in Chinese speakers, has been previously linked with an enhanced right AF and reduced left-ward lateralisation[47,49], which is suggested to be due to more experience in auditory-motor adjustments associated with using auditory feedback[49]. Similarly, Halwani and colleagues[50] had previously shown an enhanced right AF in musicians compared to non-musicians.

The auditory cortex and surrounding areas in the lateral temporal lobe, particularly in the left hemisphere, play a critical role in language processing for both tonal and non-tonal languages. However, studies using functional magnetic resonance imaging (fMRI) and magnetoencephalography (MEG) have confirmed that tonal language speakers recruit not only the left-hemisphere language areas (such as Broca's and Wernicke's areas) but also right-hemisphere regions involved in pitch perception[12–15]. This bilateral activation reflects the brain's ability to integrate pitch (a key feature of tonal languages) with linguistic information. Even during a task of tone production, Chang et al.[11] found, with higher lexical tone processing requirements, enhanced activation in the right frontal and temporal language areas, which were linked by the authors with enhanced interactions through the right AF. In a recent study, Qi et al.[14] studied native English-speaking adults enroled in a one-month Chinese course, and found that white-matter structure in the right hemisphere predicted success at early stages of Chinese language learning. Consistent with this, Loui et al.[51] found a relationship between the volume of the right AF and behavioural performances on a pitch-based artificial grammar learning task. Therefore, the right hemisphere was associated with early success at learning Chinese.

Thus, our findings are consistent with previous findings and support the stance that bilingualism per se is not linked specifically to a rightward asymmetry of the AF, but rather, significant experience and exposure to tonal languages and music are. The fact that the group differences we observed for volume were not observed in terms of microstructural properties (i.e., fractional anisotropy (FA)) is also consistent with the study from Vaquero et al.[47] suggesting that the effect on macrostructure only may be due to changes in terms of myelin packing properties (thickness, morphology) of the AF without affecting microstructural properties[47,52].

Owing to our examination of multiple different early-language learner groups, we were able to determine that smaller tract volumes in the left hemisphere and a more bilateral organization are not attributable to bilingualism or exposure to several languages in general but are specific to having been exposed to a tonal language very early in life. We show that early exposure to a tonal language, such as learning Mandarin from birth, is associated with structural brain changes in the AF and SLF III. This structural adaptation reflects the brain's early optimization of the connectivity between regions involved in pitch perception, auditory processing, and speech production. This indicates that specific linguistic properties, such as lexical tone, engage and shape neural circuits in unique ways. These findings align with theories of experience-dependent plasticity and suggest that the neural architecture supporting language processing is sensitive to language-specific features, such as pitch and tone[53]. The results tend to support the idea of experience-induced plasticity shaped by tonal language input. These findings are consistent with a study[54] which found that the languages we are exposed to and use leave a trace that can be observed at the level of the structural connectivity in the brain, and that stronger bilateral representation is associated with the processing of tonal languages. Importantly, the current findings suggest that very early language experiences, such as those observed in the IA group, may modulate the trajectory of bilingualism-induced neuroplasticity, particularly in white matter. While the Dynamic Restructuring Model (DRM) conceptualizes bilingual brain adaptations as a non-linear, experience-driven process[55,56], our data point to the possibility that brief but early exposure to a language, even if not maintained, may set into motion neurodevelopmental processes that mirror or even amplify those predicted by the DRM. Our findings underscore the need to incorporate timing, continuity, and context of language exposure, not merely its duration or proficiency level, into updated frameworks of bilingual neuroplasticity. Moreover, the results argue for a more nuanced understanding of bilingualism in neurocognitive research, one that takes into account the typological and phonological characteristics of the languages involved. Broad conclusions about the neural effects of bilingualism must therefore be interpreted in the context of specific language experiences.

Interestingly, we show that the neural architecture resulting from early but discontinued exposure to tonal languages is not overwritten. The similarity in white matter tracts between IA and Mandarin speakers clearly illustrates that early experiences influence neural structure and this is maintained in the brain. This extends the findings of the unconscious influence of neural processing by Pierce et al.[35,36] and shows that early

acquired information has an enduring influence on the brain at the structural level. Prior studies have shown that exposure to specific linguistic features (e.g., phonemes, syntax, prosody) can shape cortical and subcortical networks involved in speech and language processing[37,57]. Our results extend this work by demonstrating that even brief exposure to a tonal language in infancy can leave long-lasting structural imprints detectable in adulthood, independent of continued language use or proficiency.

Finally, although we also observed differences in the SLF III, there is something special about the AF, especially in terms of how the tract macrostructure develops with age. For the SLF III, there was no Group by Age or Group by French Exposure Duration interaction, which supports that those relationships are specific to the AF. One possible explanation for this is the natural way in which the different brain areas develop; it has been previously shown that the temporo-frontal white matter connections make the AF mature later than the fronto-parietal connections and thus they may be more sensitive to the effects of experience in early childhood[3,6,58–60]. Moreover, a more protracted white matter tract development has been linked with better receptive language skills or cognitive abilities in general[27,61].

We cannot rule out that some of the effects observed in the current study may be attributable to ethnicity or genetically influenced brain differences between Chinese and Caucasian participants. However, several studies support our interpretation that the observed differences are likely attributable to language experience. Language learning-related changes in the brain have been observed in European or American participants learning tonal languages; for example, changes in white matter tracts connecting language areas have been linked to better performance in learning a new tonal language[62]. Moreover, Crinion et al.[63] observed structural changes in the right temporal lobe that were associated with learning a new tonal language in both Chinese and European subjects learning Chinese as a second language, leading the authors to conclude that changes were due to language experience and not dependent on ethnicity. Tang et al.[64] also support the idea that the different cognitive demands involved in processing Chinese language, as well as cultural differences, likely influence the structural differences in the temporal lobe that they observed between Caucasian and Chinese individuals. Similarly, Kochunov and colleagues[65] argue that anatomical differences between Caucasian and Chinese individuals in brain areas commonly found to show functional differences between Chinese and English language tasks are more likely to be due to language experience than to ethnicity. The fact that we observed structural white matter differences specifically in the left language-dominant hemisphere and that there was some specificity in the left AF also favours an effect of experience. This is consistent with dual-stream and bilateral models of language processing[66] and further supports the idea that tonal languages may recruit right-hemisphere resources traditionally associated with prosody and music[67]. This finding underscores the importance of considering both hemispheres in models of language processing, particularly for languages that make extensive use of pitch. Of note, our structural analysis did not reveal any significant difference in terms of total intracranial volume between our group of Caucasian individuals and our group of Asian individuals. Nonetheless, future work is needed to specify whether and/or which brain differences between Caucasian and tonal-language-speaking Asians are due to genetics or language experience.

It is widely accepted that the brain's language network is primarily shaped by language experience and the specific characteristics and associated cognitive demands of the language spoken or exposed to[68]. Specifically, early learning of tonal languages appears to shape the brain in profound and unique ways, particularly by enhancing auditory processing, pitch discrimination, and influencing neural plasticity[48]. The brain's adaptation to the demands of tonal language learning highlights the remarkable flexibility of neural circuits and provides valuable insights into how the brain organizes language and auditory functions. The adaptations in the AF seen in tonal language speakers may influence their ability to learn additional languages, particularly other tonal languages--a question for future research. In addition, future research should clarify the mechanisms underlying these enduring structural changes. For example, longitudinal studies would potentially provide more direct evidence of relationships between early experience and later brain organization.

The brain exhibits tremendous plasticity and a pronounced ability to adapt to its environment. Part of its ability to adapt and learn, however, may involve retaining information potentially valuable for later use. Alternatively, it may be the case that the benefits of establishing and maintaining representations that help the learner make sense of complex sounds in their native language during infancy outweigh the costs of not maintaining those representations and instead having to continually reinterpret which sounds are relevant and why. Previous research supports the present results by demonstrating that a lost first language leaves functional and behavioural traces[34,69–72] and the current study now provides structural evidence that early-formed language representations are maintained in the brain even if exposure to that language is discontinued.

As one of the unique capacities of the human brain, the nature of language learning and representation has been a major interest of scientific research for decades. There are more than 7000 different spoken languages used for communication in the world today[73,74]. By exploring the brain networks subserving universal properties across languages and specific differences associated with different languages or language typologies, such research helps address essential questions about brain organization, maturation, and development. Our results emphasize the need for theoretical models that incorporate both the timing and the qualitative features of early language input, as well as the enduring nature of early-acquired neural patterns. Future studies examining how structural brain differences in international adoptees relate to enhanced tone discrimination or tonal language learning abilities would help clarify and extend the current findings and hypotheses.

Methods

Participants

Using MR diffusion imaging tractography, we dissected the left and right arcuate fasciculus (AF) and superior longitudinal fasciculus (SLF) to compare four groups of highly proficient French speakers (9–36 years at the time of testing). All groups had been exposed to French and used French daily, but, critical to this study, the earliest language-learning experiences of each group differed. One group had been exposed to French from birth ($n = 33$, 3 males, age range: 10–33 years). Although some of the French speakers reported some exposure to other languages, they were French-dominant L1 speakers, and importantly, none had previously been exposed to tonal languages. A second group was exposed to Mandarin from birth (learning French between the ages of 0 and 16 years) and maintained both languages at the time of testing (Mandarin–French bilinguals (Mandarin group), $n = 26$, all female, age range 9–28 years). The third group comprised individuals adopted from China (international adoptees (IA)) into French-speaking families in the first two years of life (between 3 and 25 months; $n = 36$, 1 male, age range 10–36 years). They had been exposed to Mandarin from birth and began learning French at the time of adoption, with discontinued exposure to their birth language at the time of adoption. Like the Mandarin-French bilingual group, the IA group experienced a delay before French onset, but this group is of particular interest since they had had exclusive exposure to French since the time of adoption, with no exposure to Mandarin. Because of this unique situation, we were able to examine the influence of early, but discontinued, language experience on neural organization. The participants of the IA, the French, and the Mandarin groups were from two different studies: 45 participants (21 IA, 12 Mandarin, 12 French; age 9-17) were from Pierce et al.[35] and 50 participants (15 IA, 14 Mandarin, 21 French; age 19–36) were newly recruited. In addition, to control for the specific languages spoken, we included a fourth group: 25 participants who were English-French bilinguals with no exposure to tonal languages (3 males, age range 18–25 years). To ensure the two bilingual groups were comparable, we conducted independent-samples t-tests to compare the Mandarin and E/F bilingual groups and found no significant differences between groups in terms of age of second language acquisition (E/F Bilinguals: range = 0–12 years old, $M = 5.1$; Mandarin: range= 0–16

years old, $M = 3.98$, t-test: $t = -1.111$, $p = .273$) and the number of years of experience with the second language (E/F Bilinguals: range = 10–20 years, $M = 16.6$; Mandarin: range= 8–23 years, $M = 15.0$, t-test: $t = -1.419$, $p = 0.164$). Adult bilinguals completed a questionnaire where they self-reported that they attained proficiency for reading, writing, speaking, and listening in each of their languages (average individual scores between 4.24 and 7 out of 7. For younger Mandarin-French bilinguals, their parents reported that they all used both Mandarin and French every day. Moreover, all participants lived and were tested in Montreal and all used French in their day-to-day lives for school and/or work.

All participants were right-handed individuals recruited in Montreal. Most IAs were female because of the adoption policies from China enacted in the early 1990's and thus were matched based on sex with the other groups. All participants demonstrated levels of cognitive functioning that were within the normal range as determined by study-specific measures. A subset of participants from each group (20 French, 13 IA, 14 Mandarin, 25 E/F bilinguals) completed the Matrix Reasoning subtest of the Wechsler's Adult Intelligence Scale IV; for those participants, the scaled scores on the Matrix Reasoning task did not differ between groups (ANOVA: $F = 1.935$, $p = .132$). The remaining participants from the study, including participants younger than 18 years old, did not complete the Matrix subtest. However, the three groups in that study were matched on a nonverbal IQ task, the spatial block tapping task, and on the sentence repetition task, and they all performed in the normal range, as reported by Pierce et al.[36]. Participants provided informed consent and all included datasets were from studies approved by the Research Ethics Committee of the Montreal Neurological Institute (MNI). All ethical regulations relevant to human research participants were followed.

Data acquisition and preprocessing

Magnetic resonance imaging (MRI) data were acquired at the McConnell Brain Imaging Centre at the MNI using a 32-channel head coil, with all scanning acquired on Siemens 3 T scanners. Subset 1 were participants from Pierce et al.[35] were scanned on a Siemens 3 T Tim Trio scanner with diffusion MRI acquisition (64 slices, TR = 8300 ms, TE = 88 ms, 2 mm^3 voxels, slice thickness of 2 mm, directions = 99, b = 1000 s/mm^2) and a T1-weighted gradient-echo structural scan (MPRAGE: 176 slices, TR = 2300 ms, TE = 2.98 ms); Subset 2 were scanned on a Siemens 3 T Prisma Fit scanner (diffusion weighted acquisition: 78 slices, TR = 3,200 ms, TE = 71 ms, 2 mm^3 voxels, slice thickness of 2 mm, directions = 100, $b1$ = 350 s/mm^2, $b2$ = 1250 s/mm^2, $b3$ = 2500 s/mm^2; T1 MPRAGE acquisition: 192 slices, TR = 2300 ms, TE = 2.96 ms). The English-French bilingual participants were included from another dataset (Kousaie et al.[75]) and matched for these analyses. They were scanned with the same 3 T Tim Trio Siemens scanner as the Pierce et al.[35] study, using a 32-channel head coil with the following scanning parameters: diffusion-weighted MRI data; 73 slices, TR = 10,000 ms, TE = 90 ms, 2 mm^3 voxels, slice thickness of 2 mm, directions = 64, b = 1000 s/mm^2, MPRAGE T1-weighted structural sequence: 192 slices, TR = 2300 ms, TE = 2.98 ms. To address differences between participants and studies, data were transformed into MNI space and "study" was included as a covariate in statistical analyses to control for the different scanning parameters across participant subsets, as further outlined below.

Image preprocessing

The TractoFlow pipeline[76–78] was used to pre-process the T1 and diffusion-weighted images. The 14 default steps used to process the diffusion weighted images included: denoising using the MRtrix3 dwidenoise tool[79], brain extraction and magnetic field susceptibility artefacts and eddy-currents deformation correction using the FSL package[80,81]. Dipy[82] was used for fibre orientation distribution functions (fODF) reconstruction as well as for DTI and fODF metrics extraction. The processing of the T1-weighted images included denoising and resampling with Dipy[82]. ANTs was used for N4 bias correction, cropping and normalisation, brain extraction and registration of the T1 image to the DWI space[83] and tissue segmentation[80]. Whole-brain tractograms were generated using anatomically-constrained particle-filtering probabilistic tractography[1,78,84] and seeding from the white matter (10 seeds per voxel). Finally, streamlines making a loop onto themselves of 300-degrees or more were removed[1]. To implement the dissection protocol proposed by Barbeau et al.[1], the ANTs transformation was applied to the streamline points to nonlinearly warp the whole brain tractogram to the MNI stereotaxic space[83,85].

Tractography

Tract reconstructions were carried out manually using Trackvis ([86], Ruopeng Wang, Van J. Wedeen, TrackVis.org, Martinos Center for Biomedical Imaging, Massachusetts General Hospital) by creating Regions of Interest (ROIs) overlaid on individual normalized anatomical images to locate accurately the gyri and sulci for each participant, using coordinates and landmarks from Barbeau et al.[1]. For all tracts, on the FA or FA-colour map, a first inclusion ROI was drawn in the coronal section immediately under the central sulcus, around the green triangle shape (Fig. 4 in yellow) to restrict the tracts to fibres running dorsally through the fronto-parietal white matter areas. For the arcuate fasciculus (AF), the two main end-point inclusion ROIs were a large frontal sphere covering most of the inferior (area 44 and 45) and middle (area 8 A and 9/46) frontal gyri (see Fig. 4, red sphere), and in the temporal lobe, a sphere covering the posterior superior temporal gyrus and sulcus and middle temporal gyrus (Fig. 4, fuchsia sphere). To make sure that the ending of the temporal part of the tract was restricted to the posterior temporal areas, an exclusion ROI plane was placed in the last coronal slice where the posterior-most part of Heschl's gyrus is visible (see reference[1] for more detailed information on ROI placement, where MNI coordinates are also provided. For the third branch of the superior longitudinal fasciculus (SLF III), the frontal inclusion sphere was the same as for the AF, but the posterior end-point was a sphere covering the supramarginal gyrus (see Fig. 4, orange sphere). Because of the shape of the supramarginal gyrus, the inclusion sphere overlapped on the angular gyrus (which is connected to the frontal lobe by a distinct tract, the SLF II), the main exclusion ROI for the SLF was another sphere covering the angular gyrus (see Fig. 4, green sphere). Because some individual variability remained after normalization to MNI space, especially in the parietal lobe, supramarginal and angular ROI spheres had to be adjusted to fit individual anatomical landmarks (sulci and gyri). On a case-by-case basis, additional exclusion ROIs were added, if necessary, when aberrant streamlines clearly diverging from the main tract were present. This whole process was done in the left and right hemispheres for each participant.

After the tract dissections were completed, we extracted a measure of white matter microstructure, Fractional Anisotropy (FA), in both hemispheres for all participants; with higher FA indicating better axonal properties (ex: diameter) or myelin content[52,87,88]. We also extracted volumetric measures, which is informative of how the tract is organized at the macrostructural level. The tracts in each hemisphere for the AF and SLF were successfully dissected in each participant, with measurements obtained for volume and FA (see Fig. 4 and Table 1). Estimated total intracranial volume (eTIV) was calculated for each participant using recon-all in FreeSurfer (Fischl, 2012, version 7.4.1; https://surfer.nmr.mgh.harvard.edu/) and executed via the CBRAIN platform[89], which leverages Compute Canada's high-performance computing (HPC) infrastructure for large-scale data analysis. T1-weighted MRI scans underwent preprocessing, including intensity normalization, non-uniformity correction, Talairach transformation to align images to the MNI305 atlas, and skull stripping, after which automated segmentation was performed[90]. eTIV was estimated using an atlas-based scaling approach, which applies the Talairach transformation matrix rather than direct voxel summation[91]. The computed eTIV values were extracted from the aseg.stats file. All preprocessing and segmentation steps were conducted using default parameters. Group averages were compared (French: $M = 125.0 \times 10^{-7}$ mm^3, $SD = 50.2 \times 10^{-7}$; IA: $M = 131.8 \times 10^{-7}$ mm^3, $SD = 33.0 \times 10^{-7}$; Mandarin: $M = 132.6 \times 10^{-7}$ mm^3, $SD = 35.8 \times 10^{-7}$; E/F Bilinguals: $M = 128.1 \times 10^{-7}$ mm^3, $SD = 44.5 \times 10^{-7}$) using an ANCOVA to control for Study and Age which revealed no

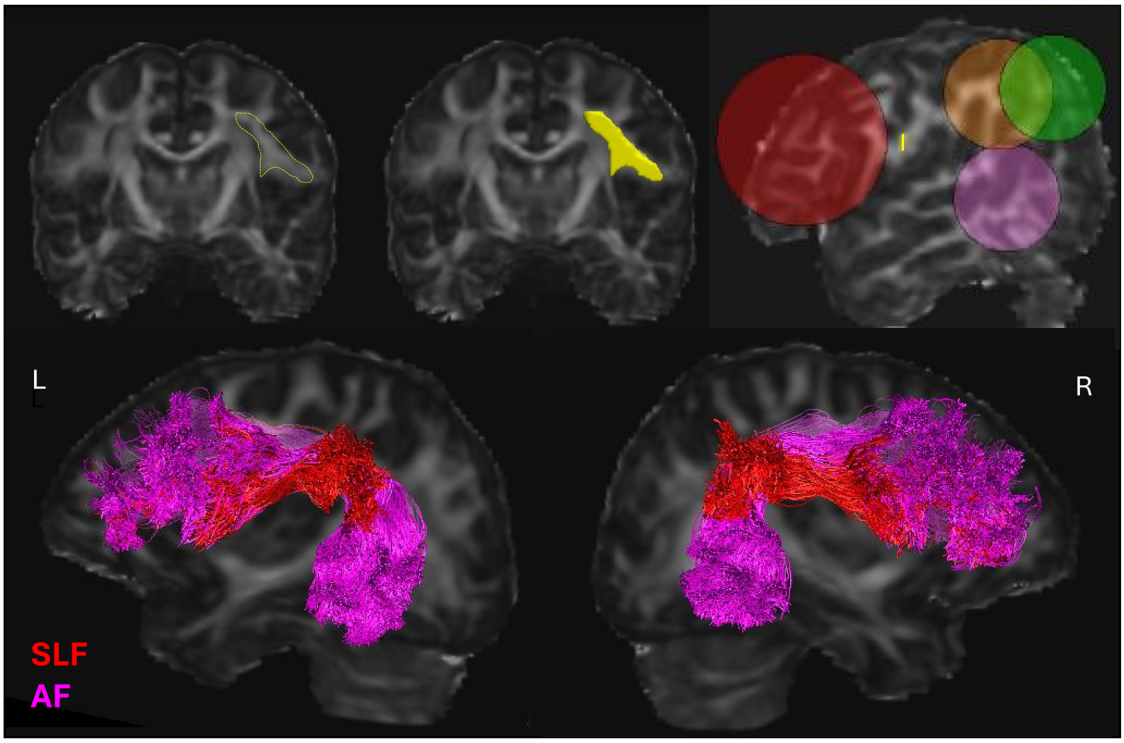

Fig. 4 | Top panel: ROI placements for the dissection of the white matter tracts. Bottom panel: Display of final dissection of the left (L) and right (R) AF and SLF in one participant.

significant difference ($F = .364$, $p = 0.696$). Even when pooling together the groups to compare Mandarin and IA to French and E/F Bilinguals, the group difference was not significant ($F = 0.727$, $p = 0.396$). Given the lack of group differences, we excluded TIV from subsequent analyses.

Statistical analysis

Results were analyzed using linear mixed effects models evaluating group differences in terms of white matter tract volumes. Effects of Hemisphere (left, right) and Tract (AF, SLF) were also investigated. A first model included Age as an interaction term with Study as a control variable and Subject as a random factor. Study was coded as a three-level dummy categorical variable; Group was coded as a four-level categorical variable with the IA group as the reference level and Hemisphere and Tract were coded as two-level categorical variables (UCLA: Statistical Consulting Group, 2011). Given that there was no Tract by Hemisphere by Age interaction, Age was included as a control variable in the final analysis (Tract Volume ~ Group * Hemisphere * Tract + Age + Study + (1 | Subject)). The same analysis was performed for tract FA values instead of volumes. Outputs from all analyses can be found as tables in the Supplementary material section.

Linear mixed effects models were implemented in RStudio Version 4.4.0[92], using the lme4 library, Version 1.1–7[93] and estimates of p-values were obtained using the lmerTest package Version 2.0–29[94].

Exploratory analyses were performed on each tract separately to further examine the Tract by Age Interactions. The presence of a Group by Age Interaction was determined using an ANCOVA and if the Interaction was significant, Pearson's correlations were used within groups to determine which group showed a significant increase in tract volume with age. Like for age, the effect of years of French exposure on tract volume was also investigated.

Data availability

The data that support the findings of this study are available as supplementary data. All other data are available from the authors upon reasonable request and with permission.

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References

1. Barbeau, E. B., Descoteaux, M. & Petrides, M. Dissociating the white matter tracts connecting the temporo-parietal cortical region with frontal cortex using diffusion tractography. *Sci. Rep.* **10**, 8186 (2020).
2. Hickok, G. & Poeppel, D. Dorsal and ventral streams: a framework for understanding aspects of the functional anatomy of language. *Cognition* **92**, 67–99 (2004).
3. Budisavljevic, S. et al. Age-related differences and heritability of the perisylvian language networks. *J. Neurosci.* **35**, 12625–12634 (2015).
4. Gervain, J. Plasticity in early language acquisition: the effects of prenatal and early childhood experience. *Curr. Opin. Neurobiol.* **35**, 13–20 (2015).
5. Gervain, J. & Mehler, J. Speech perception and language acquisition in the first year of life. *Annu Rev. Psychol.* **61**, 191–218 (2010).
6. Tak, H. J., Kim, J. H. & Son, S. M. Developmental process of the arcuate fasciculus from infancy to adolescence: a diffusion tensor imaging study. *Neural Regen. Res.* **11**, 937–943 (2016).
7. Lenneberg, E. H. On explaining language. *Science* **164**, 635–643 (1969).
8. Gandour, J. et al. Neural circuitry underlying perception of duration depends on language experience. *Brain Lang.* **83**, 268–290 (2002).
9. Xu, Y. et al. Activation of the left planum temporale in pitch processing is shaped by language experience. *Hum. Brain Mapp.* **27**, 173–183 (2006).
10. Gandour, J. Tone dissimilarity judgments by Chinese listeners. *J. Chin. Linguist.* **12**, 235–261 (1984).
11. Chang, C. H. & Kuo, W. J. The neural substrates underlying the implementation of phonological rule in lexical tone production: An fMRI study of the Tone 3 Sandhi Phenomenon in Mandarin Chinese. *PLoS One* **11**, e0159835 (2016).
12. Gandour, J. et al. Neural correlates of segmental and tonal information in speech perception. *Hum. Brain Mapp.* **20**, 185–200 (2003).

13. Luo, H. et al. Opposite patterns of hemisphere dominance for early auditory processing of lexical tones and consonants. *Proc. Natl. Acad. Sci. USA* **103**, 19558–19563 (2006).
14. Qi, Z., Han, M., Garel, K., San Chen, E. & Gabrieli, J. D. White-matter structure in the right hemisphere predicts Mandarin Chinese learning success. *J. Neurolinguist.* **33**, 14–28 (2015).
15. Wang, X. D., Wang, M. & Chen, L. Hemispheric lateralization for early auditory processing of lexical tones: dependence on pitch level and pitch contour. *Neuropsychologia* **51**, 2238–2244 (2013).
16. Kwok, V. P. Y. et al. A meta-analytic study of the neural systems for auditory processing of lexical tones. *Front Hum. Neurosci.* **11**, 375 (2017).
17. Barbeau, E. B. et al. The importance of the dorsal branch of the arcuate fasciculus in phonological working memory. *Cereb. Cortex* **33**, 9554–9565 (2023).
18. Lopez-Barroso, D. et al. Word learning is mediated by the left arcuate fasciculus. *Proc. Natl. Acad. Sci. USA* **110**, 13168–13173 (2013).
19. Saur, D. et al. Ventral and dorsal pathways for language. *Proc. Natl. Acad. Sci. USA* **105**, 18035–18040 (2008).
20. Martin, K. C., Ketchabaw, W. T. & Turkeltaub, P. E. Plasticity of the language system in children and adults. *Handb. Clin. Neurol.* **184**, 397–414 (2022).
21. Rodriguez-Fornells, A., Cunillera, T., Mestres-Misse, A. & de Diego-Balaguer, R. Neurophysiological mechanisms involved in language learning in adults. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* **364**, 3711–3735 (2009).
22. Sander, K. et al. Frontoparietal anatomical connectivity predicts second language learning success. *Cereb. Cortex* **32**, 2602–2610 (2022).
23. Petrides, M. *Neuroanatomy of Language Regions of the Human Brain*. (Academic Press, 2014).
24. Petrides, M. *The Ventrolateral Frontal Region*. In: *Neurobiology of Language*. (Academic Press, 2016).
25. Petrides, M. *The mid-dorsolateral prefronto-parietal network and the epoptic process*. In: *Principles of Frontal Lobe Function*, *2nd Ed*., 7 (Oxford University Press, 2013).
26. Brauer, J., Anwander, A. & Friederici, A. D. Neuroanatomical prerequisites for language functions in the maturing brain. *Cereb. Cortex* **21**, 459–466 (2011).
27. Estrada, K. A. et al. Language exposure during infancy is negatively associated with white matter microstructure in the arcuate fasciculus. *Dev. Cogn. Neurosci.* **61**, 101240 (2023).
28. Lebel, C. et al. Diffusion tensor imaging of white matter tract evolution over the lifespan. *Neuroimage* **60**, 340–352 (2012).
29. Stephens, R. L. et al. White matter development from birth to 6 years of age: a longitudinal study. *Cereb. Cortex* **30**, 6152–6168 (2020).
30. Giedd, J. N. et al. Brain development during childhood and adolescence: a longitudinal MRI study. *Nat. Neurosci.* **2**, 861–863 (1999).
31. Tierney, A. L. & Nelson, C. A. 3rd Brain development and the role of experience in the early years. *Zero Three* **30**, 9–13 (2009).
32. Kuhl, P. K., Conboy, B. T., Padden, D., Nelson, T. & Pruitt, J. Early speech perception and later language development: Implications for the" critical period. *Lang. Learn. Dev.* **1**, 237–264 (2005).
33. Werker, J. F. & Hensch, T. K. Critical periods in speech perception: new directions. *Annu. Rev. Psychol.* **66**, 173–196 (2015).
34. Werker, J. F. & Tees, R. C. Speech perception as a window for understanding plasticity and commitment in language systems of the brain. *Dev. Psychobiol.: J. Int. Soc. Dev. Psychobiol.* **46**, 233–251 (2005).
35. Pierce, L. J., Klein, D., Chen, J. K., Delcenserie, A. & Genesee, F. Mapping the unconscious maintenance of a lost first language. *Proc. Natl. Acad. Sci. USA* **111**, 17314–17319 (2014).
36. Pierce, L. J., Chen, J. K., Delcenserie, A., Genesee, F. & Klein, D. Past experience shapes ongoing neural patterns for language. *Nat. Commun.* **6**, 10073 (2015).
37. Kuhl, P. K. Brain mechanisms in early language acquisition. *Neuron* **67**, 713–727 (2010).
38. Werker, J. F. & Hensch, T. K. Critical periods in speech perception: new directions. *Annu Rev. Psychol.* **66**, 173–196 (2015).
39. Knudsen, E. I. Capacity for plasticity in the adult owl auditory system expanded by juvenile experience. *Science* **279**, 1531–1533 (1998).
40. Hofer, S. B., Mrsic-Flogel, T. D., Bonhoeffer, T. & Hubener, M. Prior experience enhances plasticity in adult visual cortex. *Nat. Neurosci.* **9**, 127–132 (2006).
41. Bjork, R. A. & Bjork, E. L. In *Memory and Society* 109-133 (Psychology Press, 2013).
42. Pallier, C. et al. Brain imaging of language plasticity in adopted adults: Can a second language replace the first?. *Cereb. Cortex* **13**, 155–161 (2003).
43. Penfield, W. & Roberts, L. *Speech and brain mechanisms*. Vol. 62 (Princeton University Press, 2014).
44. Fernandez-Miranda, J. C. et al. Asymmetry, connectivity, and segmentation of the arcuate fascicle in the human brain. *Brain Struct. Funct.* **220**, 1665–1680 (2015).
45. Matsumoto, R. et al. Hemispheric asymmetry of the arcuate fasciculus: a preliminary diffusion tensor tractography study in patients with unilateral language dominance defined by Wada test. *J. Neurol.* **255**, 1703–1711 (2008).
46. Vassal, F. et al. Combined DTI Tractography and Functional MRI study of the language connectome in healthy volunteers: extensive mapping of white matter fascicles and cortical activations. *PLoS One* **11**, e0152614 (2016).
47. Vaquero, L., Rousseau, P. N., Vozian, D., Klein, D. & Penhune, V. What you learn & when you learn it: Impact of early bilingual & music experience on the structural characteristics of auditory-motor pathways. *Neuroimage* **213**, 116689 (2020).
48. Hämäläinen, S., Sairanen, V., Leminen, A. & Lehtonen, M. Bilingualism modulates the white matter structure of language-related pathways. *Neuroimage* **152**, 249–257 (2017).
49. Cui, A. X. et al. Musical sophistication and multilingualism: effects on arcuate fasciculus characteristics. *Hum. Brain Mapp.* **45**, e70035 (2024).
50. Halwani, G. F., Loui, P., Ruber, T. & Schlaug, G. Effects of practice and experience on the arcuate fasciculus: comparing singers, instrumentalists, and non-musicians. *Front Psychol.* **2**, 156 (2011).
51. Loui, P., Li, H. C. & Schlaug, G. White matter integrity in right hemisphere predicts pitch-related grammar learning. *Neuroimage* **55**, 500–507 (2011).
52. Zatorre, R. J., Fields, R. D. & Johansen-Berg, H. Plasticity in gray and white: neuroimaging changes in brain structure during learning. *Nat. Neurosci.* **15**, 528–536 (2012).
53. Gandour, J. et al. Hemispheric roles in the perception of speech prosody. *Neuroimage* **23**, 344–357 (2004).
54. Goucha, T., Anwander, A., Adamson, H. & Friederici, A. D. Native language leaves distinctive traces in brain connections. *bioRxiv*, 2022.2007.2030.501987 (2022).
55. Pliatsikas, C. Understanding structural plasticity in the bilingual brain: The Dynamic Restructuring Model. *Bilingualism: Lang. Cogn.* **23**, 459–471 (2020).
56. Pliatsikas, C. et al. The effect of bilingualism on brain development from early childhood to young adulthood. *Brain Struct. Funct.* **225**, 2131–2152 (2020).
57. Newport, E. L. Maturational constraints on language learning. *Cogn. Sci.* **14**, 11–28 (1990).
58. Fibla, L. et al. Language Exposure and Brain Myelination in Early Development. *J. Neurosci.* **43**, 4279–4290 (2023).

59. Reynolds, J. E., Grohs, M. N., Dewey, D. & Lebel, C. Global and regional white matter development in early childhood. *Neuroimage* **196**, 49–58 (2019).
60. Romeo, R. R. et al. Language exposure relates to structural neural connectivity in childhood. *J. Neurosci.* **38**, 7870–7877 (2018).
61. Girault, J. B. et al. White matter microstructural development and cognitive ability in the first 2 years of life. *Hum. Brain Mapp.* **40**, 1195–1210 (2019).
62. Schlegel, A. A., Rudelson, J. J. & Tse, P. U. White matter structure changes as adults learn a second language. *J. Cogn. Neurosci.* **24**, 1664–1670 (2012).
63. Crinion, J. T. et al. Neuroanatomical markers of speaking Chinese. *Hum. Brain Mapp.* **30**, 4108–4115 (2009).
64. Tang, Y. et al. Brain structure differences between Chinese and Caucasian cohorts: A comprehensive morphometry study. *Hum. Brain Mapp.* **39**, 2147–2155 (2018).
65. Kochunov, P. et al. Localized morphological brain differences between English-speaking Caucasians and Chinese-speaking Asians: new evidence of anatomical plasticity. *Neuroreport* **14**, 961–964 (2003).
66. Hickok, G. & Poeppel, D. The cortical organization of speech processing. *Nat. Rev. Neurosci.* **8**, 393–402 (2007).
67. Zatorre, R. J., Belin, P. & Penhune, V. B. Structure and function of auditory cortex: music and speech. *Trends Cogn. Sci.* **6**, 37–46 (2002).
68. Wei, X. et al. Native language differences in the structural connectome of the human brain. *Neuroimage* **270**, 119955 (2023).
69. Au, T. K., Knightly, L. M., Jun, S.-A. & Oh, J. S. Overhearing a language during childhood. *Psychol. Sci.* **13**, 238–243 (2002).
70. Hyltenstam, K., Bylund, E., Abrahamsson, N. & Park, H.-S. Dominant-language replacement: The case of international adoptees. *Bilin.: Lang. Cognition* **12**, 121–140 (2009).
71. Oh, J. S., Au, T. K.-F. & Jun, S.-A. Early childhood language memory in the speech perception of international adoptees. *J. Child Lang.* **37**, 1123–1132 (2010).
72. Oh, J. S., Jun, S.-A., Knightly, L. M. & Au, T. K. Holding on to childhood language memory. *Cognition* **86**, B53–B64 (2003).
73. Eberhard, D. M., Simons, G., F. & Fenning, C. D. (SIL International, Dallas, Texas, 2024).
74. Leben, W. R. (Oxford University Press, 2018).
75. Kousaie, S., Chai, X. J., Sander, K. M. & Klein, D. Simultaneous learning of two languages from birth positively impacts intrinsic functional connectivity and cognitive control. *Brain Cogn.* **117**, 49–56 (2017).
76. Di Tommaso, P. et al. Nextflow enables reproducible computational workflows. *Nat. Biotechnol.* **35**, 316–319 (2017).
77. Kurtzer, G. M., Sochat, V. & Bauer, M. W. Singularity: Scientific containers for mobility of compute. *PloS One* **12**, e0177459 (2017).
78. Theaud, G. et al. TractoFlow: A robust, efficient and reproducible diffusion MRI pipeline leveraging Nextflow & Singularity. *Neuroimage* **218**, 116889 (2020).
79. Tournier, J.-D. et al. MRtrix3: A fast, flexible and open software framework for medical image processing and visualisation. *Neuroimage* **202**, 116137 (2019).
80. Jenkinson, M., Beckmann, C. F., Behrens, T. E., Woolrich, M. W. & Smith, S. M. Fsl. *Neuroimage* **62**, 782–790 (2012).
81. Smith, S. M. Fast robust automated brain extraction. *Hum. Brain Mapp.* **17**, 143–155 (2002).
82. Garyfallidis, E. et al. Dipy, a library for the analysis of diffusion MRI data. *Front. Neuroinform.* **8**, 8 (2014).
83. Avants, B. B., Tustison, N. & Song, G. Advanced normalization tools (ANTS). *Insight J.* **2**, 1–35 (2009).
84. Girard, G., Whittingstall, K., Deriche, R. & Descoteaux, M. Towards quantitative connectivity analysis: reducing tractography biases. *Neuroimage* **98**, 266–278 (2014).
85. Greene, C., Cieslak, M. & Grafton, S. T. Effect of different spatial normalization approaches on tractography and structural brain networks. *Netw. Neurosci.* **2**, 362–380 (2018).
86. Wang, R., Benner, T., Sorensen, A. G. & Wedeen, V. J. In *Proc. Int. Soc. Mag. Reson. Med*. (Berlin).
87. Schmierer, K. et al. Diffusion tensor imaging of post mortem multiple sclerosis brain. *Neuroimage* **35**, 467–477 (2007).
88. Beaulieu, C. The basis of anisotropic water diffusion in the nervous system - a technical review. *NMR Biomed.* **15**, 435–455 (2002).
89. Sherif, T. et al. CBRAIN: a web-based, distributed computing platform for collaborative neuroimaging research. *Front Neuroinform* **8**, 54 (2014).
90. Fischl, B. et al. Whole brain segmentation: automated labeling of neuroanatomical structures in the human brain. *Neuron* **33**, 341–355 (2002).
91. Buckner, R. L. et al. A unified approach for morphometric and functional data analysis in young, old, and demented adults using automated atlas-based head size normalization: reliability and validation against manual measurement of total intracranial volume. *Neuroimage* **23**, 724–738 (2004).
92. Posit team. *RStudio: Integrated Development Environment for R*, http://www.posit.co/ (2024).
93. Bates, D., Mächler, M., Bolker, B. & Walker, S. Fitting linear mixed-effects models using lme4. *J. Stat. Softw.* **67**, 1–48 (2015).
94. Kuznetsova, A., Brockhoff, P. B. & Christensen, R. H. B. lmerTest Package: tests in linear mixed effects models. *J. Stat. Softw.* **82**, 1–26 (2017).

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Competing interests

The authors declare no competing interests.

Additional information

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